Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________

Warm-Up: Max or Min?

Example (Warm-Up: Max or Min?):
For $y = 4(x-2)^2 + 7$, since $a = 4 > 0$, the function has a minimum value of 7.

Directions: State whether each function has a maximum or minimum, and give that value.

1. $y = -2(x+1)^2 + 10$

2. $y = 5(x-3)^2 - 6$

3. $y = -(x-4)^2 + 1$

4. $y = \frac{1}{2}(x+2)^2 - 8$

Find the Vertex from Standard Form

Example (Find the Vertex from Standard Form):
For $y = -3x^2 + 12x + 2$, find $x = -\frac{b}{2a} = -\frac{12}{2(-3)} = 2$, then $y = -3(2)^2 + 12(2) + 2 = 14$. Maximum value is 14.

Directions: Find the x-coordinate of the vertex, then the max/min value for each function.

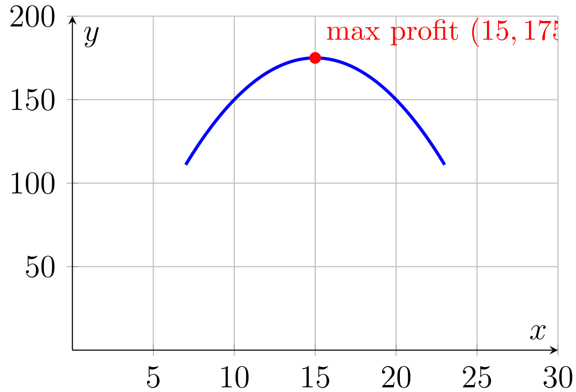
5. $y = -x^2 + 6x - 3$

6. $y = 2x^2 - 8x + 5$

7. $y = -4x^2 + 16x + 1$

Real-World Applications

Example (Real-World Applications):
A vendor's profit is $P(x) = -x^2 + 30x - 50$, where x is the price in dollars. The vertex occurs at $x = -\frac{30}{-2} = 15$, giving $P(15) = 175$. The maximum profit is $175 dollars at a price of 15$.



Profit model $P(x) = -x^2 + 30x - 50$ reaching a maximum of \{}175*at*x=15$.

Directions: Set up the vertex for each model, find the max/min value, and interpret the result in a full sentence with units.

8. A rocket's height is $h(t) = -16t^2 + 96t + 10$ (feet, t in seconds). Find the maximum height and when it occurs.

9. A rectangular pen against a barn uses 60 ft of fencing on 3 sides: $A(x) = -2x^2 + 60x$. Find the maximum area and the width that produces it.

Challenge (same sheet):

10. A company's cost to produce x items is $C(x) = 2x^2 - 40x + 300$. Find the minimum cost and the number of items that produce it.

Answer Key — fully worked

Procedure: Determine whether the function opens up (minimum) or down (maximum) from the sign of a.; Find the x-coordinate of the vertex using $x = h$ (vertex form) or $x = -\dfrac{b}{2a}$ (standard form).; Substitute that x-value back into the function to find the max/min value, k.; Interpret the result in the context of the problem, including correct units.; State your final answer as a full sentence: the maximum/minimum [quantity] is [value], occurring at [x-value]..

1. Maximum value of 10 (since $a = -2 < 0$).

2. Minimum value of -6 (since $a = 5 > 0$).

3. Maximum value of 1 (since $a = -1 < 0$).

4. Minimum value of -8 (since $a = \frac{1}{2} > 0$).

5. $x = -\frac{6}{-2} = 3$; $y = -9 + 18 - 3 = 6$. Maximum value is 6.

6. $x = -\frac{-8}{4} = 2$; $y = 8 - 16 + 5 = -3$. Minimum value is -3.

7. $x = -\frac{16}{-8} = 2$; $y = -16 + 32 + 1 = 17$. Maximum value is 17.

8. $t = -\frac{96}{-32} = 3$; $h(3) = -144 + 288 + 10 = 154$. The maximum height is 154 feet, occurring at 3 seconds.

9. $x = -\frac{60}{-4} = 15$; $A(15) = -450 + 900 = 450$. The maximum area is 450 square feet, occurring when the width is 15 feet.

10. $C(x) = 2x^2 - 40x + 300$ has $a = 2 > 0$, so it opens up and has a minimum. Vertex x-coordinate: $x = -\frac{-40}{2(2)} = \frac{40}{4} = 10$. Minimum cost: $C(10) = 2(10)^2 - 40(10) + 300 = 200 - 400 + 300 = 100$. The minimum cost is \$100, achieved when 10 items are produced.